# 23. Navigation without internet — map, compass, sun, stars and sea

Oracle OS guide. Based on US Army FM/TC 3-25.26 (Map Reading and Land Navigation), Ordnance Survey "Map reading made easy" and Bowditch "American Practical Navigator" (in the library **with illustrations**). Oracle shows your GPS position and coordinates without signal (Map tab) and Organic Maps routes offline; this guide is for when the phone fails.

## Reading the map

- **Scale**: 1:25 000 → 4 cm = 1 km; 1:50 000 → 2 cm = 1 km.
- **Contour lines** join equal heights. Close together = steep; far apart = gentle. "V"s pointing uphill = valley/stream; pointing downhill = spur/ridge.
- **Grid reference** (OS maps in the UK, also UTM/MGRS): numbers along the bottom first (**"along the corridor, then up the stairs"**), then the side. 6 figures = 100 m precision.
- **Set the map**: turn it until map features line up with the ground (or map north with compass north).

## Compass

- **Take a bearing from the map**: lay the compass edge between where you are and where you're going (direction arrow toward the target); turn the housing until its lines are parallel to the map's north–south lines, N to north; read the bearing.
- **Magnetic declination**: difference between grid north and magnetic north. In the UK and Portugal it is small (near 0°) this decade, but check the map margin — elsewhere it can be 10–20°.
- **Walk a bearing**: hold the compass in front, turn your body until the red needle sits in the housing arrow; pick a landmark in that direction, walk to it, repeat.
- **Back bearing** (return): add or subtract 180°.
- **Fix your position (resection)**: take bearings to two or three known visible features (peaks, masts), convert to back bearings, draw the lines on the map; they cross where you are.
- Keep the compass away from metal, magnets, phones and cars.

## Techniques good navigators use

- **Handrails**: follow linear features (river, wall, road, power line).
- **Catching feature**: something that tells you you've overshot (a river, a road).
- **Aiming off**: to reach a bridge on a river, deliberately aim to one side; when you hit the river you know which way to turn.
- **Pacing**: count each time the same foot lands; an adult takes ~60–70 double paces per 100 m on the flat (calibrate yours). More uphill, in rough ground or at night.
- **Timing**: walking ~4–5 km/h on the flat + 1 minute per 10 m of climb (Naismith's rule).

## Without a compass

- **Sun**: rises roughly east and sets roughly west (further north-east/north-west in summer). At solar noon it is due south (northern hemisphere) or north (southern hemisphere).
- **Shadow stick**: plant a vertical stick; mark the shadow tip; wait 15–20 min and mark again. The line from first to second mark runs **west to east**.
- **Analogue watch** (northern hemisphere): point the hour hand at the sun; south is halfway between it and 12 (use 1 o'clock during summer time). Southern hemisphere: point 12 at the sun; north is halfway between 12 and the hour hand.
- **Pole Star** (northern hemisphere): in the Plough (Big Dipper) extend the line through the two end "pointer" stars ~5 times; it reaches Polaris, which marks north and doesn't move.
- **Southern Cross** (southern hemisphere): extend the long axis ~4.5 times; drop vertically to the horizon: south.
- Moss, anthills and "the side of trees" are **not reliable**.

## Coordinates and GPS

- Formats: decimal degrees (55.8642, -4.2518), degrees-minutes-seconds, UTM/MGRS, OS grid. Always state the format when passing coordinates by radio.
- Write down coordinates of shelter, water and meeting points on paper.
- Phone GPS works without signal; save battery with airplane mode and location on.

## At sea and on rivers

- **Nautical charts**: depths in metres (soundings), hazards, lights with their character (e.g. Fl(2) 10s = 2 flashes every 10 s).
- **IALA A buoyage** (Europe, Africa, most of Asia and Oceania): **entering** harbour, **red to port** (left), green to starboard. **IALA B** (Americas, Japan, Korea, Philippines): the reverse ("red right returning").
- **Cardinal marks** (yellow and black, two cones): pass on the named side. North: cones up, black over yellow; South: cones down, yellow over black; East: cones base to base, black-yellow-black; West: cones point to point, yellow-black-yellow.
- **Tides — rule of twelfths**: over ~6 hours the water rises/falls 1/12, 2/12, 3/12, 3/12, 2/12, 1/12 of the range each hour. Streams are strongest in the middle hours.
- **Dead reckoning**: distance = speed × time; log course, time and speed every hour on paper.
- **Collision rules**: overtaking vessel keeps clear; head-on both turn to starboard (right); crossing — give way to the vessel on your right; power gives way to sail (except in narrow channels). Lights: red port, green starboard, white stern and masthead.
- Rivers: run the outside of bends (deeper) but watch for fallen trees; never approach weirs and dams from upstream.

## Plan before you go

Tell someone your route, return time and who is going. Carry a paper map, compass, whistle, torch, spare clothing, water and food.